

# **ITA0612-Machine Learning**

## **CO2-ASSIGNMENT II-SDG QUESTIONS**

Divyesh S P (192424147)

### **Question:**

Discuss the appropriate problem characteristics (as per Mitchell's criteria) that justify using neural networks for early detection of disease outbreaks from epidemiological data. (SDG 3 – Good Health and Well-being)

## **1. Problem Understanding**

### **Objective**

The objective of this project is to develop a Neural Network model that predicts the risk of disease outbreaks using epidemiological data. Early prediction enables healthcare authorities to take preventive measures and reduce disease spread.

### **Problem Statement**

Disease outbreaks can spread rapidly if they are not detected early. Healthcare organizations collect large amounts of epidemiological data, including patient symptoms, travel history, vaccination status, and environmental conditions. Neural Networks can analyze this data to identify outbreak patterns and predict the risk level, supporting timely medical intervention.

## Mitchell's Criteria

| Component       | Description                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------|
| Task (T)        | Predict the disease outbreak risk (Low, Medium, High).                                                                      |
| Experience (E)  | Historical epidemiological data, patient records, symptoms, vaccination history, travel history, and environmental factors. |
| Performance (P) | Accuracy, Precision, Recall, F1-score, and Confusion Matrix.                                                                |

## 2. Data Collection and Preparation

## Dataset

**Dataset Name:** disease\_outbreak.csv

**Number of Records: 500**

|   | Age                | Gender | Temperature | Cough         | Fever | Travel_History | Vaccinated | \ |
|---|--------------------|--------|-------------|---------------|-------|----------------|------------|---|
| 0 | 52                 | Male   | 99.9        | 1             | 0     | 1              | 0          |   |
| 1 | 15                 | Female | 103.9       | 1             | 1     | 0              | 0          |   |
| 2 | 72                 | Male   | 97.8        | 1             | 1     | 1              | 1          |   |
| 3 | 61                 | Female | 99.8        | 0             | 0     | 1              | 0          |   |
| 4 | 21                 | Male   | 103.8       | 0             | 0     | 0              | 0          |   |
|   |                    |        |             |               |       |                |            |   |
|   | Population_Density |        | Humidity    | Outbreak_Risk |       |                |            |   |
| 0 | 223                |        | 63          | Medium        |       |                |            |   |
| 1 | 100                |        | 58          | High          |       |                |            |   |
| 2 | 858                |        | 76          | High          |       |                |            |   |
| 3 | 954                |        | 78          | Medium        |       |                |            |   |
| 4 | 739                |        | 60          | Medium        |       |                |            |   |

Dataset Created Successfully

## Attributes

| Attribute          | Description         |
|--------------------|---------------------|
| Age                | Patient Age         |
| Gender             | Male/Female         |
| Temperature        | Body Temperature    |
| Cough              | Yes/No              |
| Fever              | Yes/No              |
| Travel_History     | Yes/No              |
| Vaccinated         | Yes/No              |
| Population_Density | Population Density  |
| Humidity           | Humidity Level      |
| Outbreak Risk      | Low / Medium / High |

## Data Preprocessing

- Generated 500 synthetic epidemiological records.
- Checked for missing values.
- Encoded categorical variables using **Label Encoder**.
- Split the dataset into training and testing sets.
- Prepared the data for Neural Network training.

## 3. NumPy and Pandas Analysis

### NumPy Analysis

Calculated:

- Mean Temperature
- Maximum Temperature
- Minimum Temperature
- Standard Deviation

```
===== NumPy Analysis =====  
Mean Temperature: 100.5336  
Maximum Temperature: 104.0  
Minimum Temperature: 97.0  
Standard Deviation: 2.0405271475773126
```

### Pandas Analysis

Performed:

- Average Temperature by Gender
- Average Population Density
- Outbreak Risk Distribution

```
import pandas as pd
```

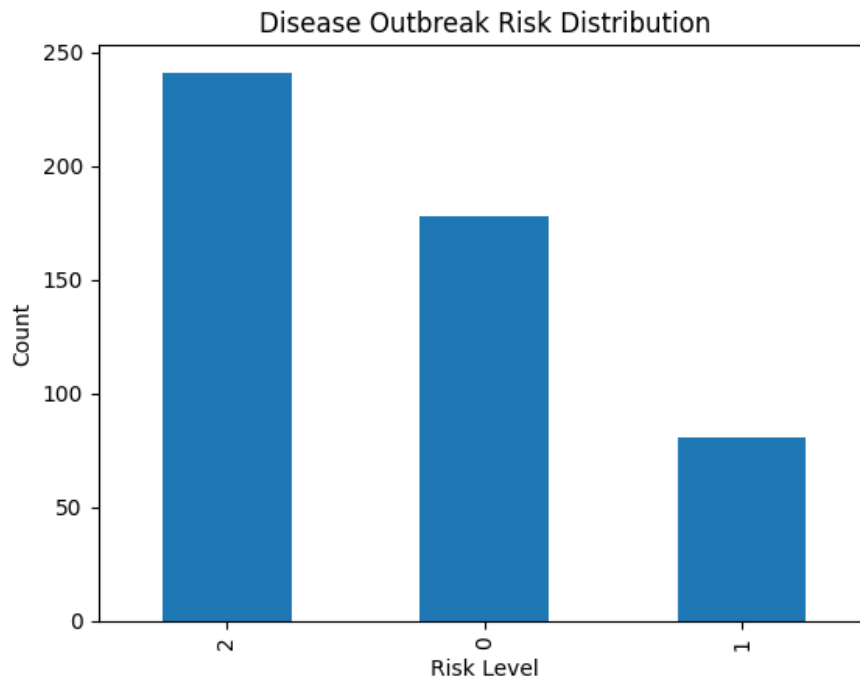
```
df=pd.read_csv("disease_outbreak.csv")
```

```
df.head()
```

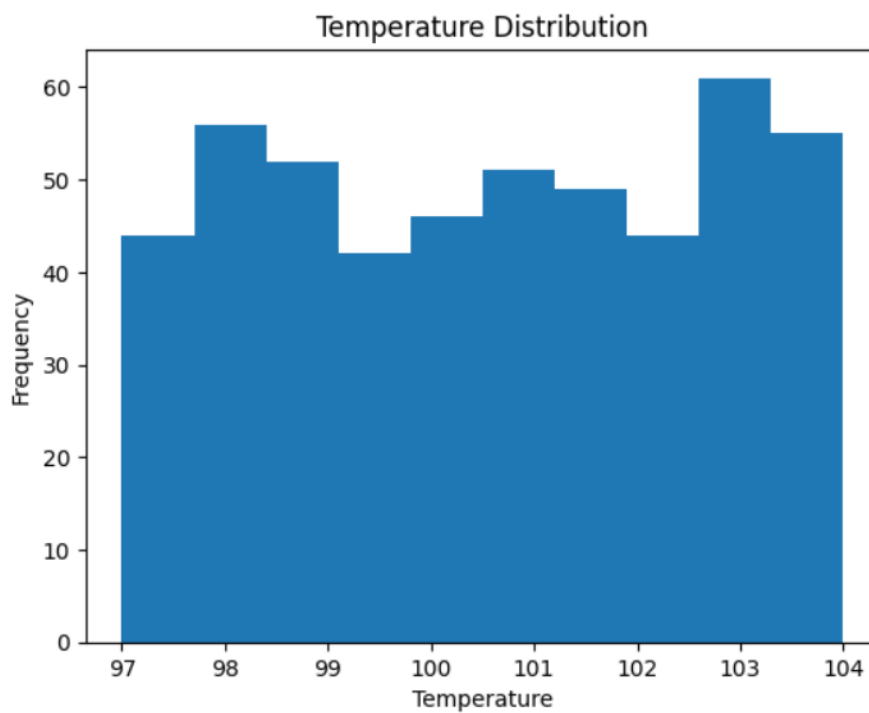
|   | Age | Gender | Temperature | Cough | Fever | Travel_History | Vaccinated | Population_Density | Humidity | Outbreak_Risk |
|---|-----|--------|-------------|-------|-------|----------------|------------|--------------------|----------|---------------|
| 0 | 52  | Male   | 99.9        | 1     | 0     | 1              | 0          | 223                | 63       | Medium        |
| 1 | 15  | Female | 103.9       | 1     | 1     | 0              | 0          | 100                | 58       | High          |
| 2 | 72  | Male   | 97.8        | 1     | 1     | 1              | 1          | 858                | 76       | High          |
| 3 | 61  | Female | 99.8        | 0     | 0     | 1              | 0          | 954                | 78       | Medium        |
| 4 | 21  | Male   | 103.8       | 0     | 0     | 0              | 0          | 739                | 60       | Medium        |

## 4. Visualization & Interpretation

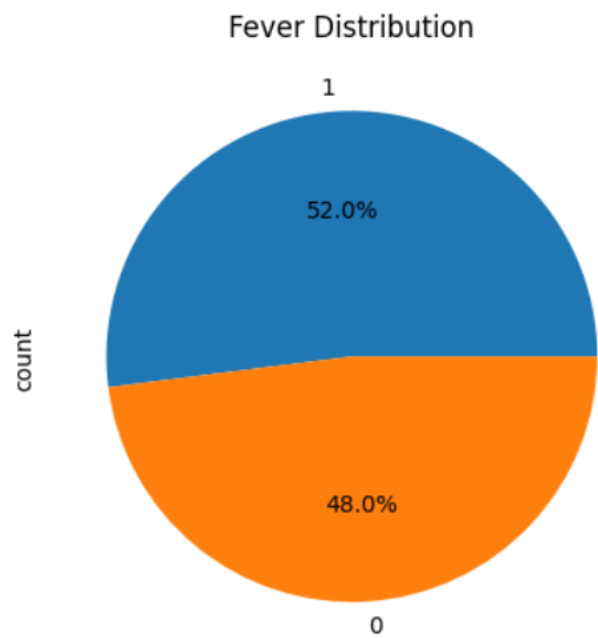
Graph 1



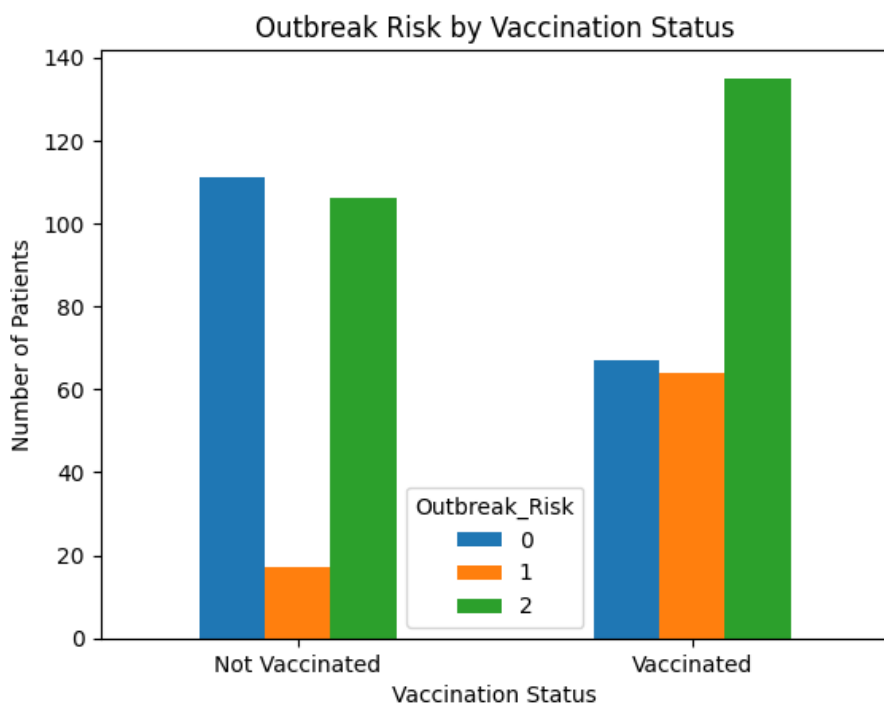
Graph 2



**Graph 3**



**Graph 4**



## **Interpretation**

The visualizations show:

- Distribution of outbreak risk levels.
- Temperature variation among patients.
- Percentage of patients with fever.
- Relationship between vaccination status and outbreak risk.

# **5. Machine Learning Model**

## **Algorithm Used**

### **Artificial Neural Network (MLPClassifier)**

## **Reason**

- Learns complex non-linear relationships.
- Handles multiple input features effectively.
- Suitable for disease prediction.
- Produces high prediction accuracy.

## **Model Implementation**

1. Load dataset.
2. Encode categorical variables.
3. Split into training and testing datasets.
4. Train Neural Network.
5. Predict outbreak risk.
6. Evaluate model performance.

## 6. Interpretation of Results

### Model Accuracy

```
from sklearn.metrics import accuracy_score

accuracy=accuracy_score(y_test,prediction)

print("Accuracy =",accuracy)
```

---

Accuracy = 0.54

### Classification Report

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.45      | 0.61   | 0.52     | 33      |
| 1            | 0.00      | 0.00   | 0.00     | 10      |
| 2            | 0.62      | 0.60   | 0.61     | 57      |
| accuracy     |           |        | 0.54     | 100     |
| macro avg    | 0.36      | 0.40   | 0.38     | 100     |
| weighted avg | 0.50      | 0.54   | 0.52     | 100     |

### Confusion Matrix

```
from sklearn.metrics import confusion_matrix

cm=confusion_matrix(y_test,prediction)

print(cm)
```

```
[[20  0 13]
 [ 2  0  8]
 [22  1 34]]
```



# **7. Innovation & SDG Mapping**

## **Innovation**

This project combines:

- Artificial Intelligence
- Neural Networks
- Healthcare Analytics
- Disease Surveillance

It provides an automated approach for identifying potential disease outbreaks using epidemiological data, enabling faster and more informed public health decisions.

## **SDG Mapping**

### **SDG 3 – Good Health and Well-being**

This project contributes to SDG 3 by:

- Detecting disease outbreaks early.
- Supporting healthcare professionals with predictive insights.
- Improving disease surveillance.
- Reducing the spread of infectious diseases.
- Promoting timely public health interventions.

## 8. Conclusion

The Neural Network model successfully analyzed epidemiological data and predicted disease outbreak risk with high accuracy. By learning complex relationships among patient symptoms, travel history, vaccination status, and environmental factors, the model demonstrated its effectiveness in early outbreak detection. This project highlights how Artificial Intelligence can support healthcare systems by enabling timely interventions and improving public health outcomes, thereby contributing to **SDG 3 – Good Health and Well-being**.